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Randomized Controlled Trial

Maternal high-dose docosahexaenoic acid supplementation and neurodevelopment at 5 Years of preterm children

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SUMMARY

Background & aims: Docosahexaenoic acid (DHA) is the most abundant omega-3 fatty acid in the brain and is accumulated by the fetal brain during the last trimester of pregnancy. Our objective was to determine whether high-dose DHA supplementation during the neonatal period, vs. placebo, improves behavioral functioning at 5 years in children born very preterm.

Methods: This is a follow-up at 5 years corrected age of a subset of children who participated in a multicenter randomized controlled trial. The participants received a high-dose DHA supplementation, or a placebo, through maternal breastmilk until 36 weeks' postmenstrual age. Primary outcome was child behavioral functioning, assessed by the Total Difficulties Score from the Strengths and Difficulties Questionnaire (SDQ). Secondary outcomes included behavioral scores from the SDQ, executive functions assessment and global developmental performance. Neurodevelopmental outcomes were assessed through interviews with parents. Mean differences between DHA and placebo groups were estimated using mixed linear models. Subgroup analyses were conducted for sex and gestational age (GA) at birth.

Results: Among 177 eligible children, 132 (74.6%) completed neurodevelopmental assessment at 5 years (DHA, N = 64, placebo, N = 68). Total Difficulties Score did not differ between the DHA and placebo groups (mean differences, −0.9 [95% confidence interval, −2.7 to 0.8], P = 0.30), nor any of the secondary outcomes. There was no significant interaction between treatment groups and sex, nor GA, for the primary outcome. However, significant interactions between treatment groups and sex or GA were found for some secondary outcomes.

Conclusions: In very preterm infants, high-dose DHA supplementation did not improve behavioral functioning at 5 years.

Clinical trial registration: ClinicalTrials.gov, NCT02371460, <https://clinicaltrials.gov/study/NCT02371460>.

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Abbreviations: ARA, Arachidonic acid; ASQ-3, Ages and Stages Questionnaire Third edition; BPD, Bronchopulmonary dysplasia; BRIEF-P, Behavior Rating Inventory of Executive Function-Preschool; CI, Confidence interval; DHA, Docosahexaenoic acid; DINO, DHA for the Improvement of Neurodevelopmental Outcome; FSIQ, Full Scale Intelligence Quotient; GA, Gestational age; GEE, Generalized estimating equations; MD, Mean difference; MOBYDICK, Maternal Omega-3 Supplementation to Reduce Bronchopulmonary Dysplasia in Very Preterm Infants; N3RO, N-3 Fatty Acids for Improvement in Respiratory Outcomes; SD, Standard deviation; SDQ, Strengths and Difficulties Questionnaire.

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1. Introduction

Children born preterm are at higher risk of adverse neurodevelopmental outcomes and neurobehavioral difficulties [1–3]. Poorer behavioral functioning during childhood is predictive of later behavior problems, with potential long-term consequences on academic performance, social interactions and family functioning [4]. With preterm birth, there is a relative deficit in docosahexaenoic acid (DHA), an omega-3 fatty acid important for brain development mainly accumulated during the last trimester of pregnancy [5,6]. Current feeding practices for preterm infants provide approximately 20 mg/kg/day of DHA, which aligns with the average level in the breastmilk of women on a Western diet, which is below the in-utero requirement of approximately 60 mg/kg/day (1 % of total fatty acids) [7,8]. This relative DHA deficit might contribute to suboptimal brain development with potential impact on neurobehavioral outcomes.

DHA, the most abundant omega-3 fatty acid in the brain, plays a structural role as a constituent of brain cell membranes, influencing membrane physical properties and thereby affecting electrical and chemical signaling in the brain. Rodent studies suggest that omega-3 deficiency during perinatal development can mimic key neuropathological and neurochemical characteristics associated with mood disorders, particularly affecting neurotransmitter systems like dopamine [9–12] and serotonin [13,14], which both play a role in behavior regulation [15,16].

Recent clinical studies of high-dose DHA supplementation in neonates born preterm suggest potential benefits on neurodevelopmental outcomes [17]. Indeed, findings from the 18-months follow-up of a randomized clinical trial suggested potential benefits from a high-dose DHA maternal supplementation for language among infants born before 27 weeks' gestation [18]. Additionally, benefits on cognitive outcomes at 5 years, without improvement of behavioral and emotional functioning, have been recently reported following direct high-dose DHA supplementation in a large cohort of very preterm neonates [19]. More data is needed to determine the role of high-dose DHA supplementation during the neonatal period to improve neurobehavioral outcomes of children born very preterm.

The current study is based on the neurodevelopmental follow-up at 5 years of a subgroup of children from the MOBYDICK trial (Maternal Omega-3 Supplementation to Reduce Bronchopulmonary Dysplasia in Very Preterm Infants; NCT02371460), a multicenter cohort of breastfed neonates born very preterm whose mothers were supplemented with DHA or placebo during the neonatal period. The objective of this study was to assess whether high-dose DHA supplementation during the neonatal period, compared to a placebo, improved childhood behavioral functioning at 5 years in children born before 29 weeks' gestation. In addition, this study explored the role of DHA supplementation on executive function and global development.

2. Material & methods

This follow-up study was conducted in children of mothers enrolled in the MOBYDICK trial [20] from one of the five sites in the province of Quebec (CHU de Québec-Université Laval, CHU de Sherbrooke, CHU Sainte-Justine, Jewish General Hospital, and McGill University Health Centre). This subset of participants of the original trial was specifically selected to participate in this follow-up study as they were eligible for a trajectory study, with a proposed eventual longer-term follow-up at school-age [21]. Participant were eligible for this follow-up study if they survived at 5 years of age, had not withdrawn from the study, and if the family initially agreed to be contacted for future research projects.

Mothers of eligible children were invited to participate in this follow-up study conducted via phone interviews with the research team using standardized neurodevelopmental questionnaires. Interviewers received standardized training to conduct interviews with the mothers when their child was aged between 57 and 66 months. All participants, researchers and clinicians responsible for the neurodevelopmental assessment were blinded to the intervention.

Trial protocol was published [21], registered at [ClinicalTrials.gov](https://clinicaltrials.gov) (NCT02371460) and approved by the research ethics board of the CHU de Québec-Université Laval and all participating sites. Furthermore, informed consent was obtained from all participating mothers before completing the interview.

2.1. Initial trial

The MOBYDICK trial is a placebo-controlled randomized clinical trial that was conducted in 16 Canadian neonatal intensive care units between June 2015 and April 2018. The original trial aimed to determine whether maternal supplementation with high-dose DHA during the neonatal period improved survival without bronchopulmonary dysplasia (BPD) in breastfed infants born before 29 weeks' gestation [20]. Eligible mothers gave birth between 23 0/7 and 28 6/7 weeks' gestation, intended to breastfeed, had no contraindications to breastfeeding and were at least 16 years old. Mothers were excluded if they had taken >250 mg/day of DHA in the 3 months prior to enrollment, or if their infant had a major congenital or chromosomal anomaly.

Mothers were randomized to receive either oral capsules of DHA (i.e. 1.2 g/day DHA to achieve 1 % of DHA in breastmilk to mimic in-utero accretion rates) or placebo capsules (a mix of corn and soy oils) within 72 h of delivery and until their infant reached 36 weeks' postmenstrual age. Randomization was conducted at the mother's level in a 1:1 ratio using computer-generated treatment allocation lists generated for each study sites by an independent statistician. The coordinating center's pharmacy coded supplementation using those randomization lists and sent blinded randomization lists and supplement bottles to each participating site's pharmacies. All participants, researchers and clinicians were blinded to group allocation.

The trial was prematurely stopped when 528 infants (461 mothers) had been enrolled due to concern for harm based on interim data analyses and also on findings from the N-3 Fatty Acids for Improvement in Respiratory Outcomes (N3RO) trial [22], published during the course of the study, raising a concern for higher risk of BPD following direct DHA supplementation.

2.2. Outcomes

All outcomes were obtained through questionnaires based on parental reports. The primary outcome was child behavioral functioning assessed with the Total Difficulties score of the Strengths and Difficulties Questionnaire (SDQ) [23]. The SDQ, a brief behavioral screening questionnaire with psychological attributes rated on a 0–10 scale, assigns a score of 0 as the optimal outcome for emotional, conduct, hyperactivity, and peer relationship domains. These four domains add up to a Total Difficulties score scored on a 0–40 scale. A Total Difficulties score of 17 or above is problematic.

Secondary outcomes included each individual scale of behavioral functioning measured by the SDQ (emotional symptoms, hyperactivity, conduct problems, peer problems, and prosocial behavior). In addition, child's executive functioning was assessed by the Behavior Rating Inventory of Executive Function–Preschool (BRIEF-P) questionnaire [24] intended for children aged 24–66 months. The BRIEF-P evaluates five clinical scales (inhibit, shift,

emotional control, working memory and plan/organize), three broader indexes (inhibitory self-control, flexibility and emergent metacognition) and a global executive composite score. Higher scores mean more symptoms of executive functions issues. Global development was assessed by the Ages and Stages Questionnaire-Third edition (ASQ-3) [25] that includes five domains of the child development (communication, gross motor, fine motor, problem solving and personal-social impairment). A score of 2 standard deviations (SD) below the average suggests that further assessment with a professional may be needed [26].

2.3. Sample size

A post-hoc sample size calculation, based on follow-up data from another cohort involved in a neonatal DHA supplementation [27], showed that a sample size of 64 participants per group provides a power of 80 %, two-tailed alpha 0.05, to detect a clinically significant difference of 0.5 SD (3.35 points) between DHA and placebo groups in the primary outcome, the Total Difficulties score of the SDQ.

2.4. Statistical analysis

Descriptive statistics, including frequencies and percentages for categorical variables and means and SD for continuous variables, were used to present baseline characteristics. For primary and secondary outcomes, mean differences (MD) between the DHA and the placebo group with 95 % confidence interval (CI) were estimated using generalized estimating equations (GEE) linear regression models, accounting for the correlation due to clustering of children from multiple births and study sites. Subgroup analyses for primary and secondary outcomes were conducted for sex and gestational age (GA) at birth (i.e. <27 vs $\geq$ 27 weeks' gestation) using GEE linear regression models with first order interaction of treatment groups with sex or GA. The Chi-squared, Fisher's exact, or Wilcoxon-Mann-Whitney tests were applied to compare baseline characteristics of the presented sample to the lost to follow-up sample. GEE log-binomial or linear regression models were used to compare participants from the province of Quebec to the original MOBYDick trial population. No imputation was performed for missing data. All statistical analyses were conducted using SAS 9.4 software (SAS Institute, NC). A significance threshold of 2-sided $P \leq 0.15$ was considered statistically significant for interaction models and $P \leq 0.05$ for all other analyses.

3. Results

3.1. Participant characteristics

Out of the 528 children randomized in the MOBYDick trial, 177 were eligible for the follow-up at 5 years at the sites in the Province of Quebec (Fig. 1). From the 177 eligible children, 134 (75.7 %) were successfully enrolled for the follow-up at 5 years, from August 2021 to July 2023. Among these participants, 64 children (53 mothers) were in the DHA group and 68 children (61 mothers) in the placebo group. Two participants were excluded from analyses as no data were collected following consent.

Most of baseline and 5 years characteristics for children in the DHA and placebo groups were similar (Table 1). However, children in the DHA group had a higher mean birth weight (947 g) compared to the placebo group (857 g). There were few significant differences among the clinical characteristics of the children from the province of Quebec in comparison with the children enrolled in the original MOBYDick trial (Supplement, eTable 1). In particular, children from the province of Quebec had a lower incidence of BPD in comparison

to the original MOBYDick cohort (30 % vs. 37 %, $P = 0.02$). Among children eligible for the 5 years follow-up, children who completed the assessment had an increased incidence of BPD and grade 3 or 4 intraventricular hemorrhage compared to those without follow-up data (Supplement, eTable 2). Additionally, the mothers of the children who completed the 5 years follow-up had a higher level of education.

3.2. Primary and secondary outcomes at 5 years

The SDQ Total Difficulties score did not differ between children in the DHA group and the placebo group (MD, -0.9 [95 % CI, -2.7 to 0.8], $P = 0.30$) (Table 2). Among the other scores from the SDQ, there was no differences between groups except for a lower peer problems score in the DHA group compared to the placebo group (MD, -0.5 [95 % CI, -0.9 to -0.03], $P = 0.04$). There were no differences between groups in any other secondary neurodevelopmental outcomes from the BRIEF-P and ASQ-3 questionnaires (Table 2).

3.3. Subgroup analyses for sex and GA at birth

Subgroup analyses stratified by sex and GA at birth for the primary and secondary outcomes are presented in Tables 3 and 4. There was no significant interaction between treatment groups and sex, nor GA, for the primary outcome.

Significant interactions between treatment and sex were observed for various secondary neurodevelopmental outcomes. Overall, a better performance was observed in girls exposed to DHA compared to placebo whereas there was no difference in boys exposed to DHA vs. placebo. More specifically, girls exposed to DHA had lower peer problems score in comparison to girls in the placebo group (MD, -0.8 [95 % CI, -1.3 to -0.2], $P = 0.006$). Additionally, girls in the DHA group had better working memory scores than girls in the placebo group (MD, -5.4 [95 % CI, -9.9 to -0.8], $P = 0.02$). Girls exposed to DHA performed better than girls in the placebo group for ASQ-3 communication (MD, 6.2 [95 % CI, 0.3 to 12.0], $P = 0.04$) and personal social scores (MD, 4.3 [95 % CI, 0.3 to 8.4], $P = 0.04$) (Table 3).

In subgroup analyses examining the effect of GA on the association between DHA and neurodevelopmental outcomes, a positive effect of DHA in neonates born <27 weeks' gestation was observed for several measures of executive functions whereas there was no difference in neonates born $\geq$ 27 weeks' gestation. In neonates born <27 weeks' gestation, DHA exposure was associated with better scores for flexibility (MD, -5.1 [95 % CI, -9.1 to -1.2], $P = 0.01$), plan/organize (MD, -7.1 [95 % CI, -13.6 to -0.6], $P = 0.03$), flexibility index (MD, -5.1 [95 % CI, -9.4 to -0.8], $P = 0.02$), metacognition index (MD, -6.8 [95 % CI, -13.1 to -0.5], $P = 0.04$) and Global composite scores (MD, -6.7 [95 % CI, -12.3 to -1.1], $P = 0.02$) (Table 4).

4. Discussion

In this 5-year follow-up of a subset of children who participated in the MOBYDick trial, a randomized controlled trial of high-dose DHA supplementation during the neonatal period in children born before 29 weeks' gestation, we found no significant difference between groups in childhood behavioral functioning, executive functions, and global neurodevelopmental performance.

Only few studies of high-dose DHA supplementation in very preterm neonates have reported on long-term neurodevelopmental outcomes. In the N3RO trial [19], a direct enteral emulsion of DHA at a dose of 60 mg/kg/day (which represents 1 % total fatty acids) among children born before 29 weeks' gestation

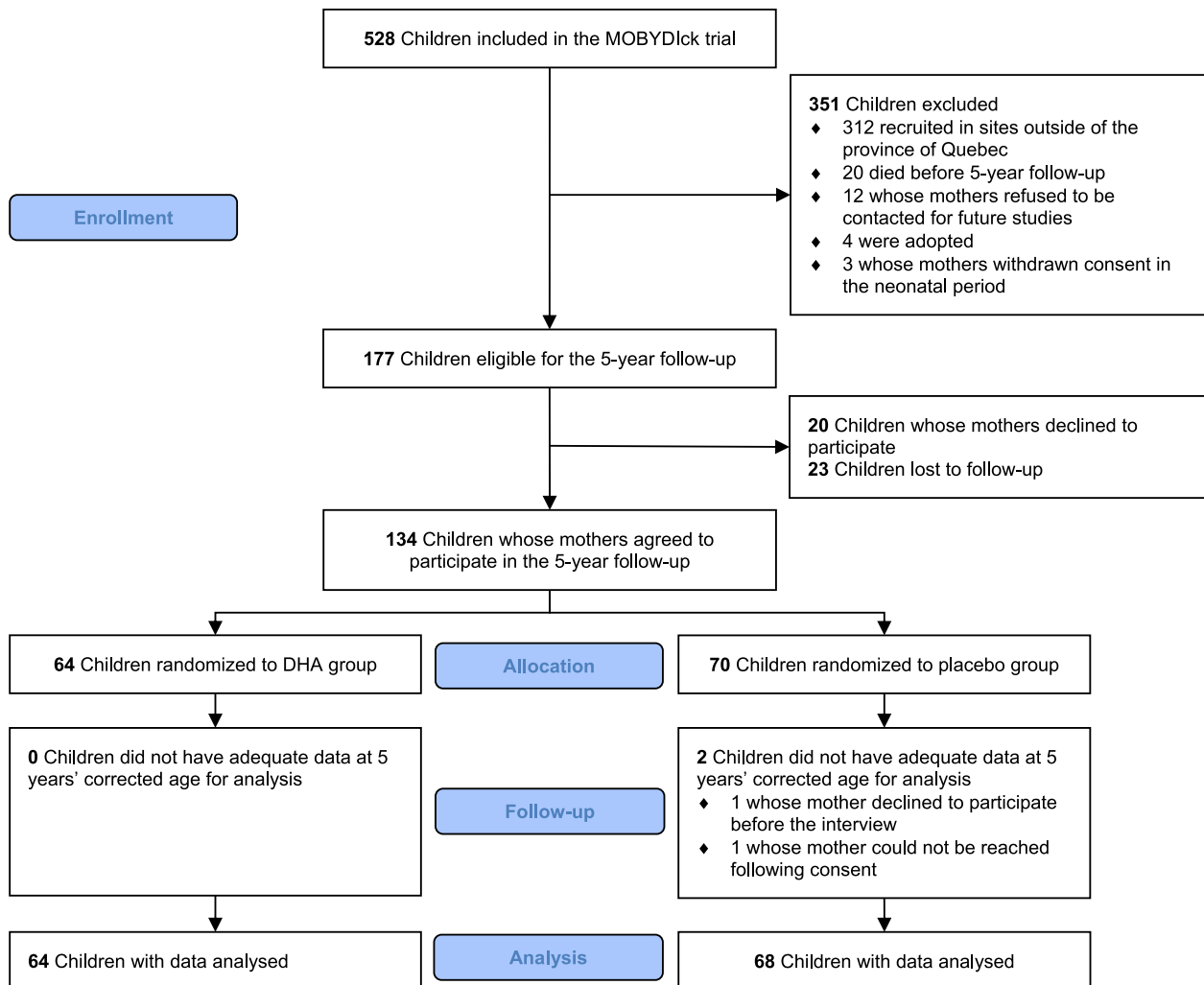


Fig. 1. Flowchart of the MOBYDlck trial's 5-year follow-up.

was associated with higher Full Scale Intelligence Quotient (FSIQ) scores at 5 years compared to a control emulsion (mean 95.4, SD 17.3 in the DHA group vs. 91.9, SD 19.1 in the placebo group, MD, 3.45 [95 % CI, 0.38 to 6.53], $P = 0.003$). However, this supplementation did not improve behavioral functioning [28]. The DHA for the Improvement of Neurodevelopmental Outcome (DINO) trial [27] examined the impact of direct enteral high-dose DHA supplementation in preterm children born before 33 weeks' gestation and revealed no significant difference in the FSIQ score at 7 years. In the DINO subgroup analysis of neonates born before 29 weeks [29], there was no difference between the DHA and the placebo groups in the FSIQ, behavioral and executive functioning. Other studies have assessed the role of DHA supplementation using various dosage of DHA in preterm children on neurodevelopment at school-age with conflicting findings. Almaas et al. [30] found no differences in cognitive performance or brain development at 8 years in very low birth weight (VLBW) infants (mean GA 28.6 weeks) given DHA (0.86 % total fatty acids) and arachidonic acid (ARA) (0.91 % total fatty acids) compared to the control group. Similarly, Isaacs et al. [31] found no significant differences at 10 years in cognitive measures between children born before 35 weeks' gestation who received long-chain polyunsaturated fatty acids-supplemented formula (0.5 % DHA) and the control group.

4.1. Observed effects in subgroups

The benefits observed in the subgroup of children born before 27 weeks' gestation exposed to DHA align with results of other previous studies [18,32]. Considering that DHA accumulates mostly during the late stages of pregnancy, it is plausible that neonates born at earlier gestational ages may exhibit a more pronounced DHA deficiency and consequently benefit preferentially from DHA supplementation.

In the subgroup of girls, DHA supplementation was associated with neurobehavioral benefits in our study. The DINO trial also suggested cognitive improvement at 18 months of age following DHA supplementation among girls only [32]. This observation may be attributed to girls' superior ability to synthesize DHA from fatty acid precursors compared to boys [33,34]. Therefore, boys may require higher DHA exposure to achieve DHA content similar to girls and detect any clinically significant effects of DHA supplementation on neurodevelopmental outcomes. However, this gender-specific effect has not been observed consistently across studies. The results from 2 large RCTs with DHA supplementation (N3RO trial [28] and DINO 7 years follow-up [27]) do not support a more beneficial effect of DHA on neurodevelopment in the female population.

Table 1
Characteristics of the mothers and their infants.

	N (%) ^a	
	DHA group	Placebo group
Mothers	(N = 53)	(N = 61)
Fatty acid levels in breastmilk 14 d postdelivery, mean (95 % CI)	[N = 48]	[N = 59]
DHA, mg/mL	0.4 (0.3; 0.4)	0.1 (0.1; 0.2)
DHA, % of total fatty acids	1.0 (0.9; 1.2)	0.4 (0.3; 0.4)
Total, mg/mL	37.8 (33.7; 41.9)	40.0 (36.1; 43.9)
Age at randomization, y, mean (SD)	31.0 (5.2)	31.8 (5.5)
Race/ethnicity ^b	[N = 52]	[N = 61]
White	34 (65)	45 (74)
Native African or African American	8 (15)	11 (18)
East or South Asian	2 (4)	0 (0)
Arab or West Asian	5 (10)	1 (2)
Latin American	3 (6)	4 (7)
Annual family income before taxes at randomization	[N = 41]	[N = 51]
<39 999\$	5 (12)	8 (16)
40 000\$–69 999\$	11 (27)	15 (29)
70 000\$–99 999\$	15 (37)	11 (22)
≥100 000\$	10 (24)	17 (33)
Highest level of education completed	[N = 52]	[N = 61]
High school or less	7 (13)	9 (15)
CEGEP, college, technical school	20 (39)	26 (43)
University	25 (48)	26 (43)
Any smoking, cigarettes or electronic cigarettes, during pregnancy	[N = 52]	[N = 61]
Any alcohol consumption during pregnancy	2 (4)	8 (13)
Any alcohol consumption during pregnancy	[N = 52]	[N = 61]
Antenatal corticosteroids	7 (14)	13 (21)
Clinical suspicion of chorioamnionitis	50 (94)	60 (98)
Received magnesium sulfate during labor	[N = 53]	[N = 60]
Cesarean delivery	4 (8)	7 (12)
Singleton pregnancy	[N = 51]	[N = 59]
	39 (77)	47 (80)
	33 (62)	44 (72)
	45 (85)	54 (89)
Infants at baseline	(N = 64)	(N = 68)
Gestational age at birth		
Weeks, mean (SD)	27.0 (1.5)	26.6 (1.6)
<27 weeks	29 (45)	35 (52)
Sex		
Male	25 (39)	34 (50)
Female	39 (61)	34 (50)
Birth weight		
Grams, mean (SD)	947 (261)	857 (231)
<10th percentile for gestational age	5 (8)	9 (13)
Age at first full enteral feeding Days, mean (SD)	[N = 54]	[N = 58]
Intravenous DHA-rich lipids before 36 weeks' postmenstrual age	26.2 (12.6)	27.8 (14.3)
Received maternal breastmilk up to 36 weeks' postmenstrual age	46 (72)	46 (68)
	62 (97)	68 (100)
Neonatal morbidities	(N = 64)	(N = 68)
Grade 3 or 4 intraventricular hemorrhage	5 (8)	10 (15)
Periventricular leukomalacia	1 (2)	3 (4)
Clinically or culture-proven sepsis	24 (38)	29 (43)
Necrotizing enterocolitis ≥ stage 2	2 (3)	1 (2)
Retinopathy of prematurity requiring treatment	2 (3)	6 (9)
Bronchopulmonary dysplasia at 36 weeks' postmenstrual age	24 (38)	18 (27)
Use of postnatal corticosteroids	21 (33)	31 (46)
Children at 5 years' follow-up	(N = 64)	(N = 68)
Most recent weight, kg, mean (SD)	[N = 59]	[N = 68]
Hospitalisation for ≥1 night since discharge	17.4 (2.6)	17.3 (2.8)
Surgical intervention	39 (61)	48 (71)
Shunt or revision for hydrocephalus	34 (53)	35 (52)
Ear, Nose and Throat surgery	1 (2)	2 (3)
Eye surgery (excluding laser for retinopathy of prematurity)	20 (31)	23 (34)
Tracheostomy	1 (2)	0 (0)
Gastrostomy	1 (2)	0 (0)
Hernia repair	4 (6)	2 (3)
Other abdominal surgery	7 (11)	3 (4)
Other surgery	1 (2)	0 (0)
Medication	9 (14)	8 (12)
Inhaled steroids	39 (61)	45 (66)
Inhaled bronchodilators	15 (23)	15 (22)
	11 (17)	14 (21)

(continued on next page)

Table 1 (continued)

	N (%) ^a	
	DHA group	Placebo group
Anticonvulsants	1 (2)	1 (2)
Anti-reflux drugs	3 (5)	0 (0)
Vitamin supplements	22 (34)	31 (46)
Other medication	18 (28)	11 (16)
Need for hearing aid or cochlear implant	[N = 64]	[N = 67]
	1 (2)	1 (2)
Visual problem	9 (14)	16 (24)
Prescription for glasses	8 (13)	14 (21)
Unilateral visual impairment (blindness)	0 (0)	1 (1)
Unknown	1 (2)	1 (1)

CI: Confidence interval; DHA: Docosahexaenoic acid; SD: Standard deviation.

^a Unless otherwise indicated.^b Self-reported by participants using multiple choice categories provided on the baseline questionnaire.

Table 2

Primary and secondary neurodevelopmental outcomes.

Outcome	Mean (95 % CI) ^a		Mean difference (95 % CI) ^a	P value ^a
	DHA group	Placebo group		
SDQ scores	(N = 64)	(N = 68)		
Total difficulties score	10.4 (9.2; 11.7)	11.4 (10.1; 12.7)	−0.9 (−2.7; 0.8)	0.30
Emotional symptoms score	2.7 (2.2; 3.3)	2.8 (2.3; 3.3)	−0.1 (−0.8; 0.6)	0.83
Peer problems score	1.1 (0.8; 1.5)	1.6 (1.3; 1.9)	−0.5 (−0.9; −0.03)	0.04
Conduct problems score	2.0 (1.6; 2.4)	2.4 (1.9; 2.9)	−0.4 (−0.9; 0.2)	0.22
Hyperactivity score	4.5 (4.0; 5.1)	4.6 (4.0; 5.2)	−0.03 (−0.8; 0.8)	0.94
Prosocial behaviour score	8.4 (8.0; 8.9)	8.7 (8.4; 9.1)	−0.3 (−0.9; 0.3)	0.35
BRIEF-P T-scores	(N = 55)	(N = 64)		
Inhibit scale	53.3 (50.6; 56.0)	53.9 (51.5; 56.3)	−0.6 (−4.3; 3.1)	0.75
Shift scale	52.9 (50.7; 55.1)	54.1 (52.2; 56.0)	−1.2 (−3.9; 1.6)	0.42
Emotional control scale	51.6 (48.1; 55.1)	50.8 (48.1; 53.6)	0.8 (−3.5; 5.0)	0.73
Working memory scale ^b	56.3 (53.8; 58.7)	59.1 (56.1; 62.1)	−2.8 (−6.5; 0.9)	0.14
Plan/organise scale ^b	52.0 (49.2; 54.7)	53.3 (50.2; 56.4)	−1.4 (−5.4; 2.7)	0.52
Behavioural regulation index	52.9 (49.8; 56.0)	52.9 (50.4; 55.5)	−0.1 (−4.1; 3.9)	0.97
Flexibility index	52.3 (49.6; 55.1)	52.7 (50.5; 54.9)	−0.4 (−3.7; 3.0)	0.83
Metacognition index ^b	54.9 (52.5; 57.3)	57.3 (54.1; 60.4)	−2.4 (−6.2; 1.4)	0.22
Global executive composite ^b	54.3 (51.6; 57.0)	55.7 (53.0; 58.4)	−1.4 (−5.2; 2.3)	0.45
ASQ-3 scores	(N = 64)	(N = 68)		
Communication score	52.9 (50.2; 55.6)	51.0 (48.2; 53.8)	1.8 (−2.1; 5.8)	0.36
Gross motor score	51.9 (49.5; 54.2)	49.7 (46.6; 52.8)	2.1 (−1.9; 6.2)	0.30
Fine motor score	48.1 (45.0; 51.2)	48.9 (45.4; 52.4)	−0.8 (−5.6; 4.0)	0.75
Problem solving score	51.7 (49.2; 54.3)	54.4 (52.5; 56.3)	−2.7 (−5.9; 0.6)	0.11
Personal social score	54.4 (52.2; 56.7)	53.3 (50.9; 55.7)	1.2 (−2.3; 4.6)	0.52

ASQ-3: Ages and Stages Questionnaire Third edition; BRIEF-P: Behavior Rating Inventory of Executive Function-Preschool; CI: Confidence interval; DHA: Docosahexaenoic acid; SDQ: Strengths and Difficulties Questionnaire.

^a The mean, mean difference and 95 % CI were derived from the GEE linear regression model including treatment main effect, site indicator and clustering of infant variables.^b Data was available for 54 infants in the DHA group and 64 infants in the placebo group.

4.2. Role of DHA in neonatal brain development: Insights from magnetic resonance imaging studies

Neonatal nutrition is a potential strategy to optimize brain development in neonates born preterm. DHA is influencing neurogenesis, synaptic membrane structure, and the efficiency of signal transmission [35]. Its accumulation in synaptic membranes supports synapse formation and enhances membrane flexibility, thereby augmenting neurotransmission and signal transduction [35].

Previous brain imaging studies have explored DHA's role in brain's development and maturation. As such, Tam et al. [36] linked higher DHA levels in children born before 32 weeks' gestation with reduced incidence of intraventricular hemorrhage, improved white matter development and larger brain volumes [37]. Moltu et al. [38] demonstrated that supplementation with ARA and DHA at doses matching estimated fetal accretion rates improves white matter maturation, as assessed by diffusion tensor imaging (DTI), compared to control treatment in infants born before 29 weeks'

gestation. However, Almaas et al. [30] study in VLBW children did not show differences in segmental brain volumes and cerebral cortex volumes at 8 years of age following DHA and ARA supplementation.

4.3. Other factors to consider

Multiple aspects related to the DHA intervention, including the mode of administration, the duration of treatment and the combination with other fatty acids such as ARA, should be considered as potentially influencing the association between DHA supplementation and neurodevelopmental outcomes. In addition to neonatal nutrition, multiple other factors should be considered. As such, children's development is a complex interplay between biological and socio-environmental factors [39]. Among the protective factors, several studies have highlighted the benefits of breastmilk in preterm neonates for long-term cognition and behavioral performance [40–43]. Our cohort's high exposure to maternal breastmilk in both groups might have influenced positively long-term

Table 3
Primary and secondary neurodevelopmental outcomes stratified by sex.

Outcome	Mean (95 % CI) ^a		Mean difference (95 % CI) ^a	P value ^a	Interaction P value ^a
	DHA group	Placebo group			
SDQ scores					
Total difficulties score					0.48
Male	11.9 (9.9; 13.9) [N = 25]	11.9 (10.0; 13.8) [N = 34]	−0.01 (−2.8; 2.7)	0.99	
Female	9.6 (8.2; 11.0) [N = 39]	10.9 (9.3; 12.6) [N = 34]	−1.3 (−3.6; 1.0)	0.26	
Emotional symptoms score					0.61
Male	2.7 (1.8; 3.6) [N = 25]	2.6 (1.9; 3.4) [N = 34]	0.1 (−1.0; 1.3)	0.86	
Female	2.7 (2.1; 3.4) [N = 39]	3.0 (2.4; 3.6) [N = 34]	−0.3 (−1.2; 0.6)	0.55	
Peer problems score					0.14
Male	1.5 (0.8; 2.2) [N = 25]	1.6 (1.2; 1.9) [N = 34]	−0.05 (−0.8; 0.8)	0.91	
Female	0.9 (0.6; 1.2) [N = 39]	1.7 (1.2; 2.1) [N = 34]	−0.8 (−1.3; −0.2)	0.006	
Conduct problems score					0.49
Male	2.4 (1.7; 3.0) [N = 25]	2.9 (2.2; 3.5) [N = 34]	−0.5 (−1.4; 0.4)	0.28	
Female	1.9 (1.5; 2.3) [N = 39]	2.0 (1.3; 2.6) [N = 34]	−0.1 (−0.8; 0.7)	0.82	
Hyperactivity score					0.48
Male	5.3 (4.4; 6.2) [N = 25]	4.9 (4.0; 5.8) [N = 34]	0.4 (−0.8; 1.7)	0.50	
Female	4.1 (3.4; 4.8) [N = 39]	4.3 (3.5; 5.1) [N = 34]	−0.2 (−1.2; 0.9)	0.74	
Prosocial behavior score					0.88
Male	8.1 (7.2; 9.0) [N = 25]	8.5 (8.0; 9.1) [N = 34]	−0.4 (−1.4; 0.6)	0.45	
Female	8.6 (8.1; 9.2) [N = 39]	8.9 (8.4; 9.4) [N = 34]	−0.3 (−1.1; 0.5)	0.45	
BRIEF-P T-scores					
Inhibit					0.78
Male	56.0 (51.0; 60.9) [N = 20]	55.4 (51.4; 59.5) [N = 31]	0.5 (−5.8; 6.8)	0.87	
Female	51.9 (49.2; 54.7) [N = 35]	52.5 (49.2; 55.8) [N = 33]	−0.6 (−5.0; 3.9)	0.81	
Flexibility					0.19
Male	54.4 (50.9; 58.0) [N = 20]	53.3 (50.7; 56.0) [N = 31]	1.1 (−3.2; 5.4)	0.61	
Female	52.0 (49.4; 54.6) [N = 35]	54.7 (52.0; 57.4) [N = 33]	−2.7 (−6.5; 1.1)	0.16	
Emotional Control					0.62
Male	57.8 (50.9; 64.7) [N = 20]	54.5 (50.0; 59.0) [N = 31]	3.3 (−4.8; 11.4)	0.43	
Female	48.5 (46.0; 51.0) [N = 35]	47.5 (44.2; 50.8) [N = 33]	1.0 (−3.2; 5.2)	0.65	
Working Memory					0.07
Male	60.8 (56.8; 64.7) [N = 19]	58.8 (53.9; 63.7) [N = 31]	1.9 (−4.3; 8.2)	0.54	
Female	53.9 (51.1; 56.6) [N = 35]	59.2 (55.3; 63.2) [N = 33]	−5.4 (−9.9; −0.8)	0.02	
Plan/Organize					0.54
Male	57.1 (52.6; 61.6) [N = 19]	56.0 (51.2; 60.8) [N = 31]	1.1 (−5.3; 7.6)	0.73	
Female	49.5 (46.5; 52.4) [N = 35]	50.9 (47.1; 54.6) [N = 33]	−1.4 (−6.1; 3.3)	0.56	
Inhibitory Self-Control Index					0.71
Male	57.4 (51.4; 63.4) [N = 20]	55.7 (51.4; 59.9) [N = 31]	1.7 (−5.5; 8.9)	0.64	
Female	50.6 (47.9; 53.3) [N = 35]	50.5 (47.1; 53.8) [N = 33]	0.1 (−4.4; 4.6)	0.96	
Flexibility Index					0.34
Male	56.5 (51.7; 61.4) [N = 20]	54.2 (50.7; 57.8) [N = 31]	2.3 (−3.5; 8.1)	0.44	
Female	50.1 (47.6; 52.7) [N = 35]	51.3 (48.3; 54.2) [N = 33]	−1.1 (−5.1; 2.9)	0.58	
Emergent Metacognition Index					0.16
Male	60.0 (56.3; 63.7) [N = 19]	58.3 (53.3; 63.3) [N = 31]	1.7 (−4.4; 7.9)	0.58	
Female	52.2 (49.6; 54.8) [N = 35]	56.2 (52.3; 60.1) [N = 33]	−4.0 (−8.5; 0.5)	0.08	
Global Executive Composite					0.24
Male	59.5 (54.8; 64.3) [N = 19]	57.4 (52.9; 61.9) [N = 31]	2.1 (−4.2; 8.5)	0.51	
Female	51.7 (49.1; 54.3) [N = 35]	54.1 (50.9; 57.4) [N = 33]	−2.5 (−6.7; 1.8)	0.25	
ASQ-3 scores					
Communication score					0.03
Male	51.0 (45.7; 56.3) [N = 25]	54.1 (50.8; 57.3) [N = 34]	−3.1 (−9.1; 2.9)	0.32	
Female	54.3 (51.4; 57.3) [N = 39]	48.2 (43.5; 52.8) [N = 34]	6.2 (0.3; 12.0)	0.04	
Gross motor score					0.87
Male	52.3 (48.3; 56.2) [N = 25]	50.4 (45.9; 54.9) [N = 34]	1.9 (−4.3; 8.0)	0.55	
Female	51.7 (48.5; 54.8) [N = 39]	49.1 (44.6; 53.6) [N = 34]	2.6 (−3.0; 8.2)	0.36	
Fine motor score					0.03
Male	45.3 (39.0; 51.6) [N = 25]	52.2 (48.2; 56.3) [N = 34]	−6.9 (−14.1; 0.2)	0.06	
Female	50.2 (47.2; 53.2) [N = 39]	45.9 (39.9; 51.8) [N = 34]	4.4 (−2.7; 11.5)	0.23	
Problem solving score					0.40
Male	51.0 (46.0; 56.1) [N = 25]	55.3 (52.6; 57.9) [N = 34]	−4.2 (−9.9; 1.5)	0.15	
Female	52.3 (49.4; 55.1) [N = 39]	53.6 (50.8; 56.3) [N = 34]	−1.3 (−5.3; 2.6)	0.51	
Personal social score					0.04
Male	50.6 (45.3; 55.9) [N = 25]	54.1 (50.6; 57.5) [N = 34]	−3.5 (−9.8; 2.9)	0.29	
Female	56.9 (55.1; 58.6) [N = 39]	52.6 (49.0; 56.1) [N = 34]	4.3 (0.3; 8.4)	0.04	

ASQ-3: Ages and Stages Questionnaire Third edition; BRIEF-P: Behavior Rating Inventory of Executive Function-Preschool; CI: Confidence interval; DHA: Docosahexaenoic acid; SDQ: Strengths and Difficulties Questionnaire.

^a The mean, mean difference and 95 % CI were derived from the GEE linear regression model including treatment main effect, sex main effect, double interaction term, site indicator and clustering of infant variables.

Table 4
Primary and secondary neurodevelopmental outcomes stratified by gestational age.

Outcome	Mean (95 % CI) ^a		Mean difference (95 % CI) ^a	P value ^a	Interaction P value ^a
	DHA group	Placebo group			
SDQ scores					
Total difficulties score					0.81
<27 weeks' gestation	10.5 (8.9; 12.0) [N = 29]	11.6 (9.8; 13.5) [N = 35]	−1.2 (−3.7; 1.4)	0.37	
≥27 to <29 weeks' gestation	10.4 (8.5; 12.3) [N = 35]	11.1 (9.5; 12.8) [N = 33]	−0.7 (−3.2; 1.8)	0.57	
Emotional symptoms score					0.24
<27 weeks' gestation	2.5 (1.8; 3.1) [N = 29]	3.0 (2.3; 3.7) [N = 35]	−0.5 (−1.5; 0.4)	0.27	
≥27 to <29 weeks' gestation	2.9 (2.1; 3.8) [N = 35]	2.6 (2.0; 3.3) [N = 33]	0.3 (−0.7; 1.4)	0.54	
Peer problems score					0.98
<27 weeks' gestation	1.4 (1.0; 1.9) [N = 29]	1.9 (1.6; 2.2) [N = 35]	−0.4 (−1.0; 0.1)	0.13	
≥27 to <29 weeks' gestation	0.8 (0.3; 1.3) [N = 35]	1.3 (0.8; 1.7) [N = 33]	−0.4 (−1.1; 0.2)	0.21	
Conduct problems score					0.86
<27 weeks' gestation	1.8 (1.2; 2.5) [N = 29]	2.3 (1.7; 2.9) [N = 35]	−0.4 (−1.3; 0.4)	0.32	
≥27 to <29 weeks' gestation	2.2 (1.7; 2.7) [N = 35]	2.5 (1.8; 3.2) [N = 33]	−0.3 (−1.1; 0.4)	0.40	
Hyperactivity score					0.50
<27 weeks' gestation	4.7 (3.9; 5.5) [N = 29]	4.5 (3.6; 5.4) [N = 35]	0.3 (−1.0; 1.5)	0.68	
≥27 to <29 weeks' gestation	4.4 (3.6; 5.1) [N = 35]	4.7 (3.9; 5.4) [N = 33]	−0.3 (−1.3; 0.7)	0.58	
Prosocial behaviour score					0.13
<27 weeks' gestation	8.8 (8.1; 9.4) [N = 29]	8.5 (8.0; 9.0) [N = 35]	0.2 (−0.6; 1.1)	0.58	
≥27 to <29 weeks' gestation	8.2 (7.5; 8.9) [N = 35]	8.9 (8.4; 9.4) [N = 33]	−0.7 (−1.6; 0.1)	0.10	
BRIEF-P T-scores					
Inhibit					0.08
<27 weeks' gestation	50.4 (47.3; 53.6) [N = 22]	54.7 (50.5; 58.9) [N = 33]	−4.3 (−9.7; 1.1)	0.12	
≥27 to <29 weeks' gestation	55.3 (51.5; 59.0) [N = 33]	53.2 (50.3; 56.0) [N = 31]	2.1 (−2.5; 6.7)	0.37	
Flexibility					0.008
<27 weeks' gestation	51.4 (48.3; 54.5) [N = 22]	56.6 (54.1; 59.1) [N = 33]	−5.1 (−9.1; −1.2)	0.01	
≥27 to <29 weeks' gestation	53.8 (50.7; 56.9) [N = 33]	51.3 (48.9; 53.8) [N = 31]	2.4 (−1.2; 6.1)	0.19	
Emotional Control					0.06
<27 weeks' gestation	48.0 (44.5; 51.6) [N = 22]	51.9 (48.3; 55.6) [N = 33]	−3.9 (−9.0; 1.2)	0.13	
≥27 to <29 weeks' gestation	54.0 (49.1; 59.0) [N = 33]	49.8 (45.5; 54.1) [N = 31]	4.2 (−2.1; 10.4)	0.19	
Working Memory					0.12
<27 weeks' gestation	55.9 (52.2; 59.5) [N = 21]	61.7 (56.9; 66.4) [N = 33]	−5.8 (−11.8; 0.2)	0.06	
≥27 to <29 weeks' gestation	56.4 (53.1; 59.7) [N = 33]	56.2 (53.1; 59.4) [N = 31]	0.2 (−4.1; 4.4)	0.94	
Plan/Organize					0.01
<27 weeks' gestation	48.8 (44.7; 52.8) [N = 21]	55.8 (50.9; 60.8) [N = 33]	−7.1 (−13.6; −0.6)	0.03	
≥27 to <29 weeks' gestation	54.0 (50.4; 57.5) [N = 33]	50.7 (47.3; 54.0) [N = 31]	3.3 (−1.5; 8.0)	0.17	
Inhibitory Self-Control Index					0.05
<27 weeks' gestation	49.4 (46.0; 52.7) [N = 22]	54.0 (49.9; 58.2) [N = 33]	−4.7 (−10.1; 0.8)	0.09	
≥27 to <29 weeks' gestation	55.3 (50.9; 59.6) [N = 33]	52.0 (48.6; 55.4) [N = 31]	3.3 (−2.1; 8.6)	0.23	
Flexibility Index					0.01
<27 weeks' gestation	49.6 (46.7; 52.6) [N = 22]	54.7 (51.7; 57.8) [N = 33]	−5.1 (−9.4; −0.8)	0.02	
≥27 to <29 weeks' gestation	54.1 (50.1; 58.0) [N = 33]	50.5 (47.3; 53.8) [N = 31]	3.5 (−1.3; 8.4)	0.15	
Emergent Metacognition Index					0.03
<27 weeks' gestation	53.2 (49.6; 56.9) [N = 21]	60.0 (54.9; 65.1) [N = 33]	−6.8 (−13.1; −0.5)	0.04	
≥27 to <29 weeks' gestation	55.8 (52.7; 58.9) [N = 33]	54.3 (51.0; 57.5) [N = 31]	1.5 (−2.6; 5.7)	0.47	
Global Executive Composite					0.01
<27 weeks' gestation	51.4 (48.2; 54.6) [N = 21]	58.1 (53.6; 62.6) [N = 33]	−6.7 (−12.3; −1.1)	0.02	
≥27 to <29 weeks' gestation	56.1 (52.3; 59.9) [N = 33]	53.3 (50.4; 56.1) [N = 31]	2.9 (−1.6; 7.3)	0.21	
ASQ-3 scores					
Communication score					0.31
<27 weeks' gestation	51.9 (47.4; 56.4) [N = 29]	48.0 (43.7; 52.4) [N = 35]	3.8 (−2.7; 10.3)	0.25	
≥27 to <29 weeks' gestation	54.0 (50.3; 57.6) [N = 35]	54.5 (51.3; 57.6) [N = 33]	−0.5 (−5.3; 4.3)	0.83	
Gross motor score					0.55
<27 weeks' gestation	51.0 (46.9; 55.0) [N = 29]	47.8 (44.1; 51.5) [N = 35]	3.2 (−2.5; 8.9)	0.27	
≥27 to <29 weeks' gestation	52.8 (49.4; 56.1) [N = 35]	52.0 (47.3; 56.7) [N = 33]	0.8 (−4.9; 6.5)	0.79	
Fine motor score					0.82
<27 weeks' gestation	47.3 (41.7; 52.9) [N = 29]	48.7 (43.4; 54.1) [N = 35]	−1.5 (−9.5; 6.6)	0.72	
≥27 to <29 weeks' gestation	48.9 (45.4; 52.3) [N = 35]	49.2 (44.4; 54.1) [N = 33]	−0.3 (−5.9; 5.2)	0.91	
Problem solving score					0.80
<27 weeks' gestation	51.2 (46.3; 56.1) [N = 29]	53.5 (50.6; 56.4) [N = 35]	−2.3 (−8.3; 3.7)	0.45	
≥27 to <29 weeks' gestation	52.2 (49.6; 54.9) [N = 35]	55.4 (52.9; 58.0) [N = 33]	−3.2 (−6.6; 0.3)	0.07	
Personal social score					0.43
<27 weeks' gestation	55.8 (52.2; 59.4) [N = 29]	53.0 (49.9; 56.2) [N = 35]	2.8 (−2.4; 7.9)	0.29	
≥27 to <29 weeks' gestation	53.3 (49.6; 57.0) [N = 35]	53.5 (49.7; 57.3) [N = 33]	−0.2 (−5.2; 4.8)	0.94	

ASQ-3: Ages and Stages Questionnaire Third edition; BRIEF-P: Behavior Rating Inventory of Executive Function-Preschool; CI: Confidence interval; DHA: Docosahexaenoic acid; SDQ: Strengths and Difficulties Questionnaire.

^a The mean, mean difference and 95 % CI were derived from the GEE linear regression model including treatment main effect, gestational age main effect, double interaction term, site indicator and clustering of infant variables.

neurodevelopmental performance. Other important neuroprotective factors include the parent-child relationship, where sensitive caregiving behaviors have been associated with better executive function and behavior [44,45]. In addition, the socioeconomic status is a key driver for the development of executive functions and cognition in children born preterm [46]. Neuroscience literature suggests that an enriched environment with optimal parental engagement influence positively postnatal brain development and neurodevelopment [47]. In our study, maternal education level and socioeconomic status were relatively high, both in the DHA and the placebo group, which might also have contributed positively to neurodevelopmental outcomes at 5 years. Furthermore, several neonatal comorbidities, including bronchopulmonary dysplasia (BPD), and the use of postnatal pharmacotherapies such as corticosteroids might contribute to adverse brain development and lead to impaired neurodevelopmental outcomes. In our study, we found a higher incidence of BPD with a higher use of postnatal corticosteroids in the DHA group in comparison to the placebo group (Table 1). However, Sullivan et al. [48] showed that the associations of DHA with BPD and the cognitive performance at 5 years of age were largely independent, suggesting that any increased BPD risk related to high-dose DHA supplementation was not associated with a reduction in the positive cognitive effects.

4.4. Strengths and limitations

Our study stands out by following a subgroup of the MOBYDick cohort from the neonatal period to 5 years, with a retention rate of 75 % among eligible children. Our study is limited by the sample size, which is based solely on the data from the 5 sites in the province of Quebec, thus potentially affecting the generalizability of our findings and limiting their interpretation. However, characteristics of the included participants were similar to those of the original MOBYDick cohort, which gives confidence to the generalizability of findings to the Canadian population. Additionally, the study involved multiple comparisons, thus increasing the risk of findings due to chance. As such, results on secondary outcomes and subgroup analyses should be considered exploratory. Nevertheless, considering the limited number of studies on high-dose DHA supplementation, particularly in a population of children born very preterm, these results are important and contribute to the advancement of current knowledge. Moreover, as our study was conducted during the COVID-19 pandemic, neurodevelopmental assessment relied on parental interviews rather than direct in-person child evaluation, which would have been more objective. Indeed, parent-completed questionnaires may introduce biases, due to their subjectivity and the risk for parents to overestimate their child's performance. However, we ensured a standardization of interviewer training. Furthermore, the neurodevelopmental assessment at 5 years relies on well-established, reliable, and validated questionnaires.

5. Conclusion

In this follow-up at 5 years of age, maternal high-dose DHA supplementation during the neonatal period to breastfed neonates born before 29 weeks' gestation was not associated with significant differences in parent-reported behavioral and neurodevelopmental outcomes at 5 years compared to placebo. In combination with the results of other trials of high-dose DHA for neurodevelopment, these findings will contribute to inform future research and clinical practices by clarifying the relevance of high-dose DHA supplementation and establishing whether subgroups of preterm neonates are more likely to benefit preferentially from supplementation, while balancing its risks and benefits.

Author contribution

Sara-Pier Paquet: Methodology, Investigation, Data curation, Formal analysis, Writing- Original draft preparation. **Etienne Pronovost:** Conceptualization, Methodology, Investigation, Data curation, Formal analysis, Supervision, Writing- Reviewing and Editing. **David Simonyan:** Methodology, Data curation, Formal analysis, Writing- Reviewing and Editing. **Georges Caouette:** Conceptualization, Investigation, Supervision, Data curation, Writing- Reviewing and Editing. **Célia Matte-Gagné:** Investigation, Supervision, Writing- Reviewing and Editing. **François Olivier:** Investigation, Supervision, Data curation, Writing- Reviewing and Editing. **Julie Bartholomew:** Investigation, Resources, Writing- Reviewing and Editing. **Alyssa Morin:** Investigation, Supervision, Writing- Reviewing and Editing. **Ibrahim Mohamed:** Conceptualization, Methodology, Writing- Reviewing and Editing. **Isabelle Marc:** Conceptualization, Methodology, Investigation, Supervision, Funding acquisition, Writing- Reviewing and Editing. **Mireille Guillot:** Conceptualization, Methodology, Investigation, Supervision, Funding acquisition, Writing- Reviewing and Editing.

All authors gave final approval of the version to be published and agreed to be accountable for all aspects of the work.

Declaration of competing interest

The authors have no conflicts of interest relevant to this article to disclose. The funders/sponsors were not involved in the design and conduct of the study, collection, management, analysis and data interpretation, preparation, review or approval of the manuscript, or in the decision to submit the manuscript for publication.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clnesp.2024.09.029>.

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